Lecture 9 Homework – Exercises

Exercise 1: Standard Normal Exponential-Shift Identity

Claim. Let $Z \sim \mathcal{N}(0,1)$. Then for all $a, b \in \mathbb{R}$,

$$\mathbb{E}\Big[e^{aZ-\frac{1}{2}a^2}\,\mathbf{1}_{\{Z>b\}}\Big] = \mathcal{N}(a-b).$$

Proof. Writing out the expectation explicitly:

$$\mathbb{E}\Big[e^{aZ-\frac{1}{2}a^2}\,\mathbf{1}_{\{Z>b\}}\Big] = \int_b^{+\infty} e^{az-\frac{1}{2}a^2}\cdot\frac{1}{\sqrt{2\pi}}\,e^{-\frac{z^2}{2}}\,dz.$$

Combining the exponentials in the integrand:

$$az - \tfrac{1}{2}a^2 - \tfrac{1}{2}z^2 = -\tfrac{1}{2}\big(z^2 - 2az + a^2\big) = -\tfrac{1}{2}(z-a)^2.$$

Therefore,

$$\mathbb{E}\Big[e^{aZ-\frac{1}{2}a^2}\,\mathbf{1}_{\{Z>b\}}\Big] = \frac{1}{\sqrt{2\pi}}\int_b^{+\infty} e^{-\frac{(z-a)^2}{2}}\,dz.$$

Substituting $u = z - a$ (so $du = dz$ and the lower limit becomes $b - a$):

$$= \frac{1}{\sqrt{2\pi}}\int_{b-a}^{+\infty} e^{-\frac{u^2}{2}}\,du = P(U > b-a), \quad U \sim \mathcal{N}(0,1).$$

Since $P(U > b-a) = \mathcal{N}(-(b-a)) = \mathcal{N}(a-b)$, we conclude

$$\mathbb{E}\Big[e^{aZ-\frac{1}{2}a^2}\,\mathbf{1}_{\{Z>b\}}\Big] = \mathcal{N}(a-b). \qquad \square$$

Exercise 2: Put-Call Parity for $t < T$

Claim. Let C_t and P_t denote the time-t prices of a European call and put with the same strike K, maturity T, on an underlying S, with risk-free rate r. Then

$$C_t - P_t = S_t - Ke^{-r(T-t)}.$$

Proof. Consider the following portfolio at time t: long one put, short one call, long one share. Its value at time t is $P_t - C_t + S_t$. At maturity T, its payoff is:

Instrument	**Initial Value**	$S_T \le K$	$S_T > K$
Long put	P_t	$K - S_T$	0
Short call	$-C_t$	0	$-(S_T - K)$
Long stock	S_t	S_T	S_T
Total	$P_t - C_t + S_t$	K	K

The portfolio pays K at maturity T in all scenarios. By no-arbitrage, its value at time t must equal the discounted value of K:

$$P_t - C_t + S_t = Ke^{-r(T-t)},$$

which rearranges to

$$C_t - P_t = S_t - Ke^{-r(T-t)}. \qquad \square$$